HW4: Byte Bistro Interactive Order System

Lecture + live demo guide

Static menu → interactive order system → JSON submission → PHP confirmation

Each part is simple. Connecting them is the challenge.

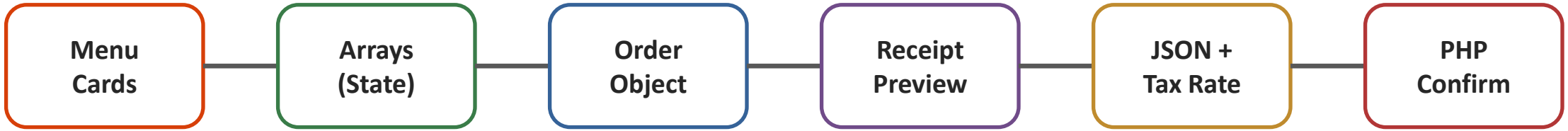
Today’s Goal

See the whole system before building the pieces

- Understand the HW4 data flow
- See how Activities 9–12 connect
- Practice debugging by tracing the order object
- Know what must match between JavaScript and PHP

The order object is the center of HW4.

Big Picture: The Full Data Flow



When something breaks, trace the data from left to right.

The Four Activities

Activity 9
Clickable menu items

Activity 10
Live Order sidebar

Activity 11
Order object + receipt preview

Activity 12
JSON submission + PHP comparison

HW4 is not four unrelated tasks. Each activity adds one layer to the same system.

Activity 9: Make the Menu Clickable

Static HTML becomes interactive UI

- Menu cards respond to clicks
- Quantity inputs appear only when needed
- data-* attributes store item information
- Console logs help verify clicks

HTML: data-* attributes

```
<div class="menu-item"
  data-category="entree"
  data-name="Tofu Tacos"
  data-price="9.50">
</div>
```

The card carries the data.

Live Demo Script: Activity 9

Click

Click a menu card

Change quantity

Click another card

Say

“This is not submitting yet.”

“We are just testing interaction.”

“The console is our developer view.”

Watch

Input appears

Buttons appear

Console logs item data

Activity 10: Store State in Arrays

The page remembers selected items

Selected item arrays

```
let entrees = [];
let beverages = [];
let desserts = [];
```

Each selected item

```
const selectedItem = {
  name: name,
  price: price,
  quantity: quantity
};
```

Activity 10 introduces state: what the current order contains.

Live Order Sidebar

State becomes visible to the user

Render arrays into sidebar

```
function updateLiveOrder() {
  updateList("entree-list", entrees);
  updateList("beverage-list", beverages);
  updateList("dessert-list", desserts);
}
```

Confirm adds or updates an array

Remove deletes from the array

updateLiveOrder() redraws the sidebar

The sidebar is not the final receipt

Live Demo Script: Activity 10

Click Tofu Tacos → enter 2 → Confirm

Point out: the item appears in Live Order

Change quantity → Confirm again

Point out: quantity updates instead of duplicating

Remove item → Live Order updates again

Key phrase: “The array changed, so the display changed.”

Activity 11: Build the Order Object

Combine menu selections + form inputs

HW3 order format

```
const order = {
  customer: ...,
  date: ...,
  city: ...,
  delivery: ...,
  tipPercent: ...,
  entrees: entrees,
  drinks: beverages,
  desserts: desserts,
  addOns: ...
};
```

This is the center of HW4

It has the same shape as HW3 orders

It connects UI data to processing functions

If this object is wrong, everything after it is wrong

Code Focus: buildOrderObject()

Reference version

```
function buildOrderObject() {
  const selectedCity = document.querySelector('input[name="city"]:checked');
  const dining =   document.querySelector('input[name="diningMethod"]:checked');

  const diningMethod = dining.value;

  const order = {
    customer: document.getElementById("customerName").value,
    date: getTodayDateString(),
   phone: //TODO
    diningMethod: diningMethod,
    delivery: diningMethod === "delivery",
    city: selectedCity.value,

    tipPercent: parseFloat(document.getElementById("tipPercent").value),
    entrees: entrees,
    drinks: beverages,
    desserts: desserts,
    addOns: getSelectedAddOns()
  };

  return order;
}
```

Code Focus: getSelectedAddOns()

Reference version

- Add-ons now use the same object structure:
  {name, price, quantity}

- HTML must include data-price:

<input type='checkbox' name='addOns' value='Avocado' data-price='1.25'>

- If add-ons appear but totals are wrong, check calculateSubtotal().

```
function getSelectedAddOns() {
  const addOnsList =
document.querySelectorAll('input[name="addOns"]
:checked');
  const addOns = [];

  addOnsList.forEach(function(item) {
    addOns.push( {
      name: item.value,
      price: parseFloat(item.dataset.price),
      quantity: 1
    });
  });

  return addOns;
}
```

Small HTML Choice: Default Radio Buttons

Use checked on one city

Use checked on one dining method

Then JavaScript can safely read .value

required is not needed when a default is selected

HTML defaults

```
<input type="radio" name="city"
       value="Corvallis" checked>

<input type="radio" name="diningMethod"
       value="pickup" checked>
```

Refactor: printReceipt()

HW3: print line by line

```
console.log("BYTE BISTRO RECEIPT");
console.log("Date: " + order.date);
```

HW4: build one string

```
let receipt = "";
receipt += "BYTE BISTRO RECEIPT\n";
receipt += "Date: " + order.date + "\n";
return receipt;
```

console.log shows output. return gives the string back.

Preview Receipt Flow

Use the returned string

menu-interact.js

```
function previewReceipt() {
  const order = buildOrderObject();

  const receipt = printReceipt(order, priceMap, taxRates);

  document.getElementById("receiptOutput").textContent = receipt;
}
```

If you forget const receipt = ..., the console may work but the page will not update.

Live Demo Script: Activity 11

Confirm a few items

Fill customer name, city, tip, and dining method

Click Preview Receipt

Point out: same receipt appears in console and on page

Change quantity or city → Preview again → values update

Key phrase: “Preview is not submission. It is the customer view.”

Activity 12: Submit JSON to PHP

Now the order leaves the browser

Hidden inputs

```
<input type="hidden" name="orderJson"
id="orderJson">
<input type="hidden" name="taxRate"
id="taxRate">
```

The user does not type into hidden inputs

JavaScript fills them before submit

orderJson carries the order object as a string

taxRate carries taxRates[order.city]

Code Focus: Submit Handler

No preventDefault()

```
document.querySelector("form").addEventListener("submit", function () {
  const order = buildOrderObject();


  // Fill hidden inputs
  document.getElementById("orderJson").value = JSON.stringify(order);
  document.getElementById("taxRate").value = taxRates[order.city];
});
```

Why target="_blank"?

```
<form id="order-form" action="menu-order.php" method="post" target="_blank">
```

Current Tab

JavaScript
receipt preview

New Tab

PHP
confirmation page

Compare the totals side-by-side.

PHP Role: Verification Page

Students use a provided menu-order.php starter

- Receives orderJson and taxRate
- Decodes JSON into PHP data
- Calculates subtotal, tax, tip, delivery, total
- Displays confirmation page

Students send correct data + style the page.

PHP uses prices in orderJson.

PHP Must Run on the Server

- PHP does not run locally.
- Upload ALL files to the ENGR server.
- Test from the server URL.
- Open your site from the server http://web.engr.oregonstate.edu/~onid/.
- Files in the SAME folder
  → use action="menu-order.php"
- Check permissions: 644

Common issues
- blank page
- 403 error
- file downloads instead of running

Debugging PHP

Optional

PHP

```
// echo "<pre>";
// print_r($order);
// echo "</pre>";
```

- Uncomment to debug
- Shows EXACT data PHP recieves
- If you use this for debugging
- The code must be commented out for the styling and format to look correct.
- Comment it out before submitting

Live Demo Script: Activity 12

Click Preview Receipt first

Then click Submit Order

Point out: the current page keeps the preview

New tab opens with menu-order.php

Compare subtotal, tax, tip, delivery, and total

Key phrase: “Same data, calculated in two places.”

Common Bug #1: Missing Preview Listener

Needed

```
document.getElementById("previewBtn")
  .addEventListener("click",
previewReceipt);
```

Symptom: Live Order works

Symptom: clicking Preview does nothing

Cause: previewReceipt() was never connected to the button

Common Bug #2: Add-ons Missing or Not Counted

HTML

```
name="addOns"
```

JavaScript

```
querySelectorAll('input[name="addOns"]:checked')
```

JavaScript

```
addItems(order.addOns, "add-on");
```

Strings must match exactly

addOns ≠ addons

Symptom: add-ons array is empty

Also include in subtotal

Common Bug #3: Console Works, DOM Does Not

Problem

```
printReceipt(order, priceMap,
taxRates);
```

Fix

```
const receipt = printReceipt(order, priceMap,
taxRates);
receiptOutput.textContent = receipt;
```

console.log shows output

return gives data back

You must store the returned string before displaying it

Common Bug #4: Menu Names Do Not Match priceMap

Menu card

```
data-name="Aloo-rithmic Curry Bowl"
```

priceMap

```
"Curry Bowl": 12.00
```

These are different keys

Prices may calculate as zero

Use exact spelling, spaces, and capitalization

Common Bug #4: Why Totals Don’t Match

Most common causes

- addOns not included
- names don’t match priceMap
- data-price missing
- Same data?
  → then check calculations

Debugging Strategy: Trace the Data

1. Does the click handler run?
2. Are arrays updated?
3. Is buildOrderObject() correct?
4. Does printReceipt() return a string?
5. Are hidden inputs filled?
6. Does PHP match preview?

Do not debug everything at once. Follow the data.

Live Demo Checklist

Confirm at least one entrée, one drink, one dessert

Choose a city with nonzero tax

Choose delivery so delivery fee appears

Choose a tip percentage

Preview, then change something and preview again

Submit and compare PHP confirmation

Why This Project Still Requires Understanding

AI can generate snippets, but it cannot automatically connect your exact files

Names, IDs, data attributes, arrays, and objects must all agree

The hard part is integration and debugging

That is real software development

The system works only when the pieces agree.

Final Takeaway

HW4 is about connections.

Menu → arrays → order object → receipt → JSON → PHP

Build one piece. Test it. Then connect the next piece.